

LAB 1: REQUIREMENTS ENGINEERING

Remote Patient Vitals Alert & Monitoring App

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Domain: Healthcare & Telemedicine

Problem Statement: #16

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall monitor patient vitals and detect critical abnormalities.	High	Heart rate >140 BPM triggers an alert within 2 seconds.	Enables quick detection of critical conditions.	Given requirement
FR-002	Functional	The system shall record patient SpO ₂ , Heart Rate, and BP.	High	All three vital readings are recorded with time.	Supports continuous patient monitoring.	Specify telemetry source
FR-003	Functional	The system shall send an alert when a vital crosses its critical threshold.	High	An alert is sent when a threshold is breached.	Ensures the caregiver is informed.	Specify alert method
FR-004	Functional	The system shall allow the caregiver to view vitals and acknowledge alerts.	High	Caregiver can view readings and acknowledge an alert.	Helps the caregiver respond quickly.	Specify acknowledgment action

FR-005	Functional	The system shall escalate an unacknowledged alert to the next caregiver.	High	Alert is escalated after the set time.	Prevents critical alerts from being missed.	Specify escalation time
NFR-001	Nonfunc.	The system shall support 500 concurrent telemetry streams with 99.99% uptime.	High	System supports 500 streams and 99.99% uptime.	Ensures reliable monitoring.	Given requirement
NFR-002	Nonfunc.	The system shall protect patient data from unauthorized access.	High	Unauthorized users cannot access patient data.	Protects sensitive patient information.	Specify security method